



HACETTEPE UNIVERSITY

Department of Nuclear Engineering

NEM 393 ENGINEERING PROJECT II

**Makeup Project: Thermal-Hydraulic Analysis of Downward Water Flow in a
Vertical Pipe Subjected to Regular Heat**

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1 ABSTRACT

This project addresses the flow of water in a vertical pipe to which a constant heat (1.57 MW/m) is applied across its entire surface. This flow is also downward. The aim of the project is to analyze the system by calculating the pressure, temperature, and phase change parameters using analytical and numerical methods. The project also aims to determine the starting point of boiling, if it occurs, and the point at which the water boils completely, if it occurs.

The analyses were performed in two different ways. In the first stage, the momentum and energy equations were solved analytically in a single step. In this approach, fluid properties were determined based only on values such as the fluid temperature at the inlet and outlet of the pipe.

In the second stage, the pipe was divided into finite control volumes, and a numerical Python code was developed using the Homogeneous Equilibrium Model (HEM). In the numerical solution, water properties such as enthalpy and specific heat were obtained using a library that can be loaded into Python and provides highly accurate and precise values, and these values were updated by iterating through each numerical step. Numerical calculations were performed using three different step intervals (1 m, 0.1 m, 0.01 m).

Despite the heat energy supplied along the pipe, the increased pressure due to gravity raises the boiling point, resulting in the fluid remaining in the liquid phase (supercooled liquid) along the pipe in both analytical and numerical results. In other words, no boiling point onset or vapor desiccation occurred. Furthermore, a perfect agreement was observed between the analytical and numerical results, demonstrating the accuracy of the result and the method.

2 INTRODUCTION

It is quite normal for a liquid to begin boiling and transition to two-phase flow depending on the conditions, but calculating this is a very difficult scenario. This is because the fluid is no longer completely liquid or completely gaseous, but in an intermediate phase; therefore, its c_p value, volume, density, and other properties change. This, of course, changes the heat transfer rate. This situation is seen in many engineering applications such as nuclear reactors or thermal power plants, heat exchangers, and industrial process lines. For the safety and efficiency of the system, these calculations must be performed with extreme accuracy.

In the system examined in this project, the flow is downward. Although there is a pressure loss due to friction in downward flow, the pressure of the system increases absolutely due to the effect of gravity. An increase in the pressure of a fluid means that its

boiling point also increases. Therefore, if there is a downward flow and this fluid is receiving heat energy from a source, it is absolutely necessary to consider not only the temperature increase but also the pressure change to analyze whether the fluid will boil or not.

The Homogeneous Equilibrium Model (HEM), which assumes that the liquid and vapor phases are in thermodynamic equilibrium, was used as the analysis method. The main objective of the project is to determine whether the fluid starts to boil under the given conditions and values, at what point boiling begins if it does, whether it boils completely, and the quality of the vapor. In addition, changes in pressure, temperature, density, and specific heat along the pipe were also analyzed.

To resolve and analyze these situations, the problem has been addressed with two different solutions:

- Analytical Solution:

The problem was solved in a single step, which utilizes the average properties of the fluid given in the problem. Of course, this solution also takes into account the increased pressure due to gravity.

- Numerical Solution:

In this method, the pipe is divided into finite volumes at specific increments. Since thermophysical properties of water, such as density and enthalpy, are known to be functions of temperature and pressure, the fluid properties are updated at each step, and calculations continue with the newly obtained values. Three different increment segments were used to ensure the accuracy of the numerical method.

The water properties used in the calculations were taken from the internationally accepted and used IAPWS-IF97 library.

3 METHODS AND CALCULATIONS

This project investigates the properties and changes in properties of water flowing down a vertical pipe. The solutions in this project are developed based on the Homogeneous Equilibrium Model (HEM). This model assumes that a fluid consisting of two different phases, water and steam, is in thermodynamic equilibrium. That is, water and steam have the same velocities and temperatures. Therefore, the state of the fluid can be determined solely by its quality.

Assumptions:

- **Steady-State:** The flow is assumed to be in a steady state; in other words, the flow parameters are constant and do not change over time.
- **1-D Flow:** It was assumed that fluid properties changed only in a dimension perpendicular to the channel axis, with no change in other axes.
- **Homogeneous Equilibrium Model**
- **Pipe Specifications:** The vertical length of the pipe is 17.2 meters, while its diameter is 0.32 meters.
- **Linear Heat Rate:** The linear heat rate along the pipe is constant and its value is 1.57 MW/m.
- **Initial Conditions:** The fluid, i.e., water, is assumed to have a pressure of 1.5 MPa and a temperature of 420 Kelvin just before entering the pipe. Under these inlet conditions, the Cp value of the water is also “4.2989 kJ/kg.K”

○ Mass Conservation:

In this system, the incoming and outgoing masses are equal, meaning the mass flow rate is conserved.

$$\frac{dm}{dz} = 0 \Rightarrow \dot{m} = \rho u A = \text{constant}$$

○ Conservation of energy:

Here, "h" represents enthalpy and "q'" represents heat flux.

$$\frac{dh}{dz} = \frac{q'}{\dot{m}}$$

○ HEM Momentum Equation:

The general HEM momentum equation for downward flow is given below. In the equation, it is assumed that the gravitational force is positive because it acts in the direction of flow.

$$\frac{dP}{dz} = \rho_{mix} g - \frac{f G^2}{2 \rho_{mix} D} - G^2 \frac{d}{dz} \left(\frac{1}{\rho_{mix}} \right)$$

Here “G = $\dot{m}$ /Area” and “f” is the friction factor.

- **Friction Factor:**

$$f = 0.316 \cdot Re^{-0.25}$$

- **The Reynolds Number:**

$$Re = \frac{GD}{\mu}$$

- **Properties of Water:**

Since the density, temperature, pressure, and enthalpy values of the liquid are interdependent, they are shown in the figure below and were obtained using IAPWS-97.

$$\rho = \rho(P, h), \quad T = T(P, h)$$

If two-phase flow occurs in the system, the mixture density and vapor dryness were calculated using the following equations.

$$x = \frac{h - h_f}{h_g - h_f}$$

$$\rho_{mix} = \left(\frac{x}{\rho_g} + \frac{1-x}{\rho_f} \right)^{-1}$$

- **Numerical analysis**

Along the pipe, the pipe was discretized using the finite difference method. That is, the length of the pipe (L) was divided into N control volumes, each with a length of Δz . In this project, the solution was obtained using 3 different Δz values.

Euler's method was used, which is an iterative method used to move from one step to the next.

Enthalpy Change/Update:

$$h_{i+1} = h_i + \frac{q' \Delta z}{\dot{m}}$$

Pressure Change/Update:

$$P_{i+1} = P_i + (\rho_i g \Delta z) - \left(\frac{f_i G^2}{2 \rho_i D} \Delta z \right)$$

4 RESULTS

This section examines the results obtained using both analytical and numerical methods with different step-by-step procedures for the described system. The aim of the project and the results discussed include investigating whether and to what extent boiling occurs along the pipe, the steam quality, the pressure change, and also the density, pressure, temperature, and specific heat values of the fluid as it passes through the pipe, based on numerical solutions.

Boiling Onset and Phase State:

The results show that although the fluid temperature increases along the pipe due to the heat source, it does not begin to boil at any point, or in other words, it does not reach its saturation temperature. This remains unchanged in both analytical and numerical solutions; both different solution methods prove that the fluid does not reach its saturation temperature and confirm the accuracy of the answer. The ultimate reason for this is that as the fluid flows downwards, the pressure increases due to gravity, resulting in an increase in the boiling point.

This directly answers another question analyzed in the project: steam quality. Steam quality refers to the ratio of the mass of vapor in the fluid to the total mass. And in the project we analyzed, since we never reached the boiling point, the fluid we have is automatically 100% liquid, meaning the steam quality is directly **zero**.

Table 1: Based on both analytical and numerical solutions, it shows the pressure, temperature, phase state, and saturation temperature of the water at the pipe outlet.

Method	Step Size	Exit Pressure (P_out)	Fluid Temperature (T_fluid)	Saturation Temperature (T_sat @ P_out)	Phase Status
Analytical Model	-	1.6503 MPa	470.95 K	476.15 K	Subcooled Liquid
Numerical Model	1.0 m	1.6507 MPa	470.94 K	476.15 K	Subcooled Liquid
Numerical Model	0.1 m	1.6505 MPa	470.94 K	476.15 K	Subcooled Liquid
Numerical Model	0.01 m	1.6505 MPa	470.94 K	476.15 K	Subcooled Liquid

The table above shows the saturation temperature at the outlet pressure, and as can be seen, the temperature did not reach this point at any stage, and boiling did not occur.

Pressure Change and Comparison of Analytical and Numerical Solutions:

As shown in Table 1, the analytical solution is extremely close to the numerical solution. The most important reason for this closeness is that the density of water was calculated based on the temperatures at the inlet and outlet points during the analytical solution.

Furthermore, the inlet pressure in the problem was 1.5 MPa. We found the outlet pressure to be 1.6505 MPa. This pressure increase is due to the flow being downward. The force of gravity caused a pressure increase of approximately 0.15 MPa.

Grid Independence Study:

To understand the accuracy of the numerical solution and obtain a more accurate and precise result, the numerical calculation was repeated with three different step intervals. (1m, 0.1m, 0.01m.) Again, as seen in Table 1, the results of the three different numerical solutions are very close to each other and are the same even when considering the significant figures chosen for the results at 0.1m and 0.01m. This closeness demonstrates the accuracy of both the analytical and numerical solutions.

In addition, we can reach the correct result even at the 1 m interval. As the step interval decreases, the accuracy of the result is expected to increase, but this also increases the computational capacity. Since efficiency is one of the most fundamental principles of engineering, choosing the correct number of steps is also important. The reason why a large step interval gives the correct result in this problem is that the fluid remains in a single phase. From this, it can be concluded that a very large step interval is not necessary in single-phase fluids, as is the case in such problems. Of course, the accuracy of the step interval will increase much more in two-phase fluids.

❖ Analysis of thermohydraulic properties of fluid using numerical solution:

Moreover to inlet and outlet analyses, the fluid's temperature, density, specific heat capacity, and pressure were analyzed in detail and displayed on a graph, depending on the path the fluid followed during flow. A step interval of 0.01 meters was chosen for this numerical analysis. Although other intervals were added to the two graphs, they were not reflected in the graphs as the results were not significantly different.

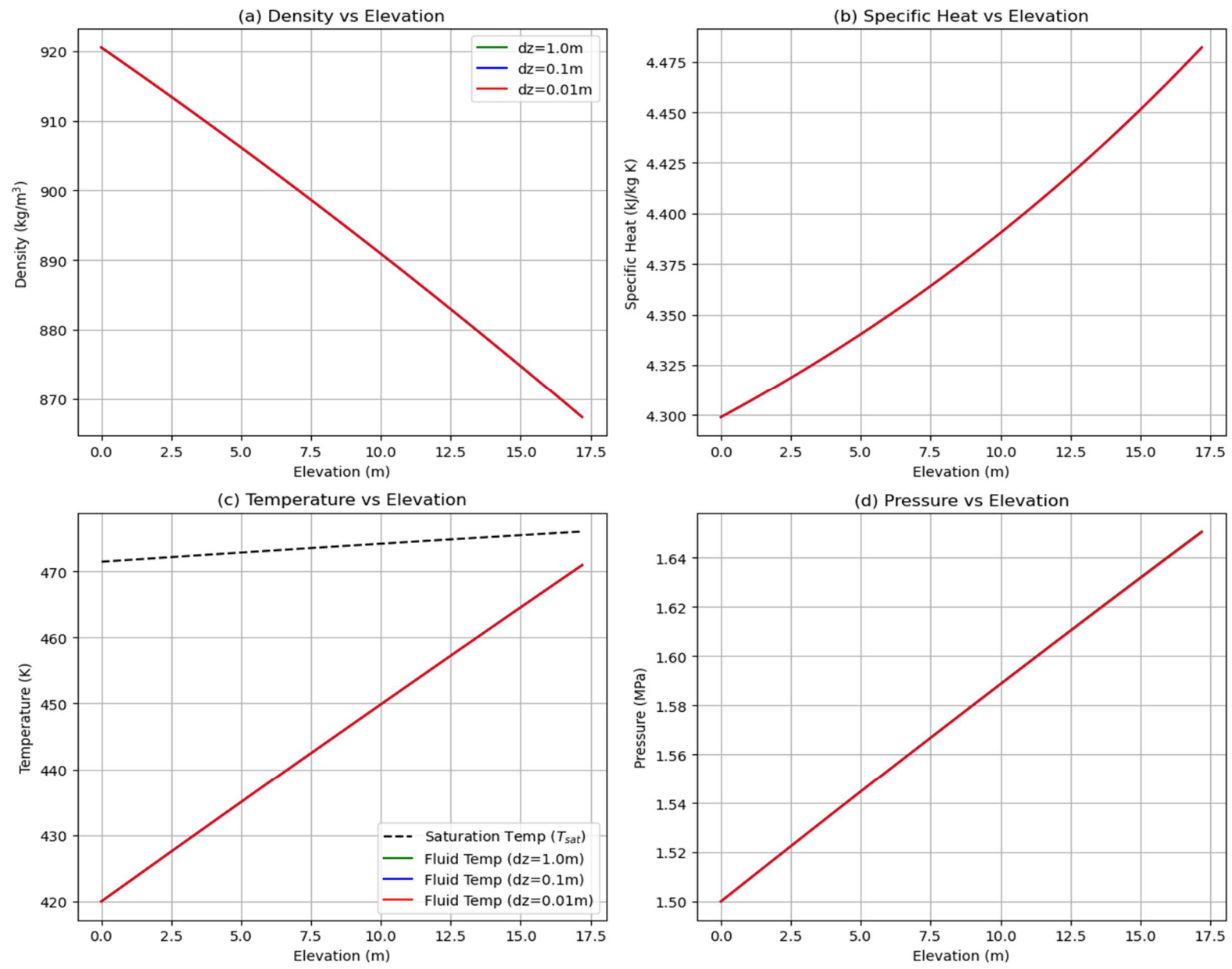


Figure 1: The graphs shows the variation in the thermal-hydraulic parameters of the fluid obtained from numerical results at 0.01 intervals along the pipe exposed to heat:

- (a) density variation due to the temperature increase caused by heating,
- (b) variation of specific heat capacity with elevation,
- (c) comparison of fluid temperature with the local saturation temperature,
- (d) pressure variation along the pipe for downward flow

Density Variation:

As shown in Figure 1a, the fluid density shows a continuous and linear decrease from an inlet value of 921 kg/m^3 to an outlet value of 867 kg/m^3 . Since the flow is downstream, the pressure increases with height. At the same time, the temperature also increases due to the applied heat input of 1.57 MW/m. The increase in temperature decreases the density, while the increase in pressure increases the density. The decrease seen in the graph proves that the

effect of temperature on density is much greater than the effect of pressure. In addition, the linear structure of this graph alone is another indication that the fluid remains in the single-phase liquid region. If a phase change such as boiling had occurred, there would be very sharp decreases in the graph.

Variation of Specific Heat:

The biggest advantage of the numerical approach is its ability to account for changes in thermophysical properties such as specific heat. Unfortunately, this is not possible in the analytical solution and is therefore not used in real-world applications. The specific heat capacity is updated at each step depending on the current pressure and temperature, leading to the graph shown in Figure 1b, which does not show a linear increase. In summary, the numerical solution, by accounting for these updates, has very accurately modelled the increase in energy storage capacity as the fluid approaches the saturation limit.

Additionally, the updated c_p value for specific intervals is given in the table below. Using the numerical solution method, it was observed that the C_p value increased by approximately 4.26% from input to output.

Table 2: This table presents the variation of specific heat capacity along the channel elevation, obtained from the numerical solution with a step size of 0.01 m.

Elevation (m)	c_p (kJ/kg·K)	Elevation (m)	c_p (kJ/kg·K)
0.00	4.2989	9.00	4.3795
1.00	4.3066	10.00	4.3904
2.00	4.3145	11.00	4.4016
3.00	4.3228	12.00	4.4133
4.00	4.3313	13.00	4.4255
5.00	4.3402	14.00	4.4381
6.00	4.3495	15.00	4.4512
7.00	4.3591	16.00	4.4649
8.00	4.3691	17.00	4.4791
		17.20	4.4820

Temperature Variation Analysis During Elevation:

The temperature variation is given in Figure 1c. Under the influence of a constant heat source, the fluid temperature increases linearly. The most important point here is the saturation temperature of the fluid, shown by the black dashed lines. As the fluid temperature gradually increases, the boiling point of the fluid also rises due to the increase in pressure in the system. This clearly explains why boiling does not begin despite the constant heat flow applied.

The fact that the temperature graphs obtained at different step intervals do not appear or, in other words, overlap in the numerical solution shows that each interval chosen in the solution of this problem correctly solves the question. This also proves the accuracy of the numerical solution.

Pressure Variation Analysis During Elevation:

Figure 1d shows the pressure distribution along the pipe. Since the flow is downstream, the pressure increases clearly and linearly. Although there is pressure loss due to friction, the linearity in the graph shows that the pressure loss due to friction is very weak compared to the pressure increase due to gravity. Furthermore, we can understand from this that the flow is single-phase. If the onset of boiling had been reached, the graph would not be so smooth; there would be a distinct break at the point where boiling begins.

5 CONCLUSION

In this project, a fluid in a vertical pipe flowing downwards and subjected to a constant heat input (1.57 MW/m) along its entire length is investigated using both analytical and numerical methods with the Homogeneous Equilibrium Model (HEM). The main objective of the analysis is to examine the changes in the fluid's (i.e., water) properties such as temperature, pressure, density, and specific heat throughout its flow. Additionally, the aim is to determine whether the regularly acting heat input initiates boiling in the fluid as it passes through the pipe, and if so, at what point this occurs, and to calculate the steam quality accordingly.

Phase State and Boiling:

Despite the applied heat input along the pipe, analyses based on the pressure-dependent increase in boiling point revealed that the liquid never reached its boiling point. From inlet to outlet, the liquid remained entirely in the supercooled liquid phase. This, of course, indicated that the vapor quality was zero.

Pressure Effect:

It was determined that the biggest factor preventing boiling is that as the temperature increases with heat input, the flow flows downwards, and the increasing pressure raises the

boiling point. The pressure increased by approximately 0.15 MPa. This is extremely critical for the safety of the systems and shows us one of the most important points.

Accuracy of Solution Methods:

Both analytical and numerical solutions were developed for the project. The analytical and numerical solutions obtained with different step intervals (1m, 0.1m, 0.01m) showed excellent agreement in terms of outlet pressure and temperature values. This agreement demonstrates the accuracy of both the methods and the selected step intervals.

Changes in Thermophysical Properties:

The biggest difference between numerical and analytical solutions is that the numerical solution can update a value like heat capacity at each step. While the fluid density decreases linearly along the flow in the pipe, the specific heat shows a different, non-linear increase. The specific heat value increased by approximately 4.26% from inlet to outlet. This result shows that the numerical solution used in the HEM model provides a more accurate answer and is one of the most critical points in the project.

In conclusion, this study clearly demonstrates the effect and importance of pressure in the flow direction, and especially gravity in the case investigated in this project, on boiling analysis. Furthermore, it has been shown that the numerical solution developed using the Homogeneous Equilibrium Model yields results with extremely high accuracy in flow problems. These results are of great importance for the effective, safe, and efficient operation of engineering systems such as thermal power plants, nuclear reactors, heat exchangers, and industrial pipelines.

6 REFERENCES

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7 APPENDIX

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from iapws import IAPWS97

# 1. INPUTS
L = 17.2
D = 0.32
Area = np.pi * (D**2) / 4.0
m_dot = 121.0
G = m_dot / Area
q_prime = 1.57e6
P_in = 1.5
T_in = 420.0
g = 9.81

# 2. ANALYTICAL CALCULATION
def solve_analytical():
    state_in = IAPWS97(P=P_in, T=T_in)
    h_in = state_in.h

    # Energy Balance
    total_heat = (q_prime * L) / 1000.0
    h_out = h_in + (total_heat / m_dot)

    # Average Properties
    state_out_est = IAPWS97(P=P_in, h=h_out)
    rho_avg = (state_in.rho + state_out_est.rho) / 2
    mu_avg = (state_in.mu + state_out_est.mu) / 2

    # Pressure Drop
    Re = (G * D) / mu_avg
    f = 0.3164 * (Re**-0.25)

    dP_grav = rho_avg * g * L
    dP_fric = f * (L/D) * (G**2) / (2 * rho_avg)

    P_out = P_in + (dP_grav - dP_fric) * 1e-6
    T_out = state_out_est.T
```

```

    return P_out, T_out

# 3. NUMERICAL CALCULATION (HEM)
def solve_hem(step_size):
    z = 0.0
    P_curr = P_in
    h_curr = IAPWS97(P=P_in, T=T_in).h

    results = []

    # Save first point
    inlet = IAPWS97(P=P_in, h=h_curr)
    results.append({'z': 0, 'P': P_curr, 'T': inlet.T, 'rho':
inlet.rho, 'cp': inlet.cp, 'x': 0})

    while z < L - 1e-6:
        dz = min(step_size, L - z)

        # Energy
        dh = (q_prime * dz / 1000.0) / m_dot
        h_next = h_curr + dh

        # Momentum
        st = IAPWS97(P=P_curr, h=h_curr)
        rho = st.rho
        mu = st.mu

        Re = (G * D) / mu
        f = 0.3164 * (Re**-0.25)

        dP_grav = rho * g * dz
        dP_fric = f * (dz/D) * (G**2) / (2*rho)
        P_next = P_curr + (dP_grav - dP_fric) * 1e-6

        z += dz
        P_curr = P_next
        h_curr = h_next

    # Data storage

```

```

        try:
            st_new = IAPWS97(P=P_curr, h=h_curr)
            x_val = st_new.x if st_new.x is not None else 0.0
            cp_val = st_new.cp if x_val == 0 else 0
            results.append({'z': z, 'P': P_curr, 'T': st_new.T, 'rho':
st_new.rho, 'cp': cp_val, 'x': x_val, 'T_sat': IAPWS97(P=P_curr, x=0).T})
        except:
            pass

    return pd.DataFrame(results)

# 4. RUN AND PRINT
P_anal, T_anal = solve_analytical()

# Print Header
print(f"{'Method':<25}      {'P_out      (MPa)':<15}      {'T_out      (K)':<15}
{'Quality'}")

# Print Analytical Result
print(f"{'Analytical':<25} {P_anal:<15.4f} {T_anal:<15.2f} {0.0:<15}")

# Run Numerical Loop
steps = [1.0, 0.1, 0.01]
hem_data = {}

for s in steps:
    df = solve_hem(s)
    hem_data[s] = df
    last = df.iloc[-1]
    print(f"{'Numerical      (dz={s})':<25}      {last['P']:<15.4f}
{last['T']:<15.2f} {last['x']:<15.5f}")

# 5. PLOTTING
fig, axes = plt.subplots(2, 2, figsize=(10, 8))

# Density
ax = axes[0,0]
for s in steps:
    ax.plot(hem_data[s]['z'], hem_data[s]['rho'],
label=f'dz={s}')
ax.set_title('(a) Density')
ax.set_ylabel('Density (kg/m3)')
ax.legend()

```

```

ax.grid(True)

# Cp
ax = axes[0,1]
for s in steps: ax.plot(hem_data[s]['z'], hem_data[s]['cp'],
label=f'dz={s}')
ax.set_title('(b) Specific Heat')
ax.set_ylabel('Cp (kJ/kgK)')
ax.grid(True)

# Temperature
ax = axes[1,0]
df_final = hem_data[0.01]
ax.plot(df_final['z'], df_final['T_sat'], 'k--', label='T_sat')
for s in steps: ax.plot(hem_data[s]['z'], hem_data[s]['T'],
label=f'dz={s}')
ax.set_title('(c) Temperature')
ax.set_ylabel('Temperature (K)')
ax.legend()
ax.grid(True)

# Pressure
ax = axes[1,1]
for s in steps: ax.plot(hem_data[s]['z'], hem_data[s]['P'],
label=f'dz={s}')
ax.set_title('(d) Pressure')
ax.set_ylabel('Pressure (MPa)')
ax.set_xlabel('Elevation (m)')
ax.grid(True)

plt.tight_layout()
plt.show()

```

```

import numpy as np
import pandas as pd
from iapws import IAPWS97

#INPUTPARAMETERS
L = 17.2
D = 0.32
Area = np.pi * (D**2) / 4.0

```

```

m_dot = 121.0
G = m_dot / Area
q_prime = 1.57e6
P_in = 1.5
T_in = 420.0
g = 9.81

#CALCULATION
def calculate_cp_profile(step_size):
    z = 0.0
    P_curr = P_in

    h_curr = IAPWS97(P=P_in, T=T_in).h
    data = []

    #0.0
    inlet_props = IAPWS97(P=P_in, h=h_curr)
    data.append({'z': 0.0, 'cp': inlet_props.cp})

    while z < L - 1e-6:
        dz = min(step_size, L - z)

        #Updating Enthalpy
        # Converiton
        dh = (q_prime / 1000.0 * dz) / m_dot
        h_next = h_curr + dh

        state = IAPWS97(P=P_curr, h=h_curr)
        rho = state.rho
        mu = state.mu

        # Friction
        Re = (G * D) / mu
        f = 0.3164 * (Re**-0.25)

        # Pressure
        dP_grav = rho * g * dz
        dP_fric = f * (dz/D) * (G**2) / (2*rho)

```



```

# Convert2
P_next = P_curr + (dP_grav - dP_fric) * 1e-6

#step
z += dz
P_curr = P_next
h_curr = h_next

try:
    current_state = IAPWS97(P=P_curr, h=h_curr)
    data.append({'z': z, 'cp': current_state.cp})
except:
    pass

return pd.DataFrame(data)

# tABLE
df = calculate_cp_profile(0.01)

#wants
targets = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,
17, 17.2]

print("\n--- CP DISTRIBUTION (dz = 0.01 m) ---")
print(f"{'Z (m)':<10} | {'cp (kJ/kg.K)':<15}")
print("-" * 30)

for t in targets:

    idx = (df['z'] - t).abs().idxmin()
    row = df.loc[idx]

    print(f"{row['z']:<10.2f} | {row['cp']:<15.4f}")

```